

THE EFFECT OF FOREIGN AID ON GENDER EQUALITY IN DEVELOPING NATIONS

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Abstract:

One of the most prominent goals of foreign aid is to increase gender equality in developing nations. We investigate if foreign aid achieves this intended national-level goal by examining impacts on a gender equality indicator, consisting of the percentage of women in the labor force, percent of women in parliament, adolescent fertility ratio, maternal mortality ratio, and female life expectancy. Using data from the World Bank from the years 2002 to 2018, we first compare regional and national differences across the indicator to account for cultural and geographical disparities. We then run a series of fixed-effects regression models with lagged variables and account for socio-economic and institutional factors such as education, governance quality, and foreign direct investment. Our results then argue that aid has little to no positive effect on gender outcomes and is largely dependent on initial government and economic country conditions used as control variables in our model.

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1 - INTRODUCTION

Gender equality joined the mainstream development discussion in the 1970s when the accumulating evidence suggested that improving gender outcomes was imperative for greater social development. Throughout this time, the United Nations and aid donors began to emphasize that a focus on women's empowerment was essential to increase the effectiveness of foreign aid (Floro, 1995). With the increased narrative of decreasing gender inequality, there was an international call to encourage gender mainstreaming and the integration of gender sensitivity in all aid projects and policies (Elgström, 2000). Perhaps most distinctly, gender equality became a recognized objective through first its prominence in the Millennium Development Goals (MDGs) and now the Sustainable Development Goals (SDGs), specifically Goal 5, which calls to "achieve gender equality and empower all women and girls" (United Nations, 2023). That said, even with a great emphasis on increasing gender mainstreaming in foreign aid, most of the existing data compiles the effect of specific aid programs and does not measure the national impact. Therefore, existing analysis mostly focuses on micro-level aid interventions, as macro-level impact tends to be much more difficult to document (Roodman, 2015). As a result, the analysis of the effect of foreign aid on gender equality at the national level remains very much in progress.

This study will contribute to the debate on the impact of foreign aid on gender equality by exploring select gender equality indicators. The study seeks to explore the following question: does an increase in the amount of foreign aid affect gender equality in developing nations?

The study uses data from the World Bank Open Data set, for developing countries from 2002 to 2018, examining how foreign aid impacts gender equality. It excludes developed countries, conflict zones, and countries with incomplete data to avoid reverse causality and ensure more accurate results.

The econometric analysis uses panel data techniques that enable us to control for country-specific impacts using fixed-effects estimations based on the construction of a principal analysis index incorporating five indicators (percent of women in the labor force, percent of women in parliament, adolescent fertility rate, maternal mortality ratio, and female life expectancy). The empirical model uses a dummy variable for a dependent variable that defines if a country ranks among the upper half or lower half of countries in terms of gender equality for a given year. It then measures the probability of moving to the upper half of countries in the rank to then derive the shift in gender equality.

The remainder of the paper is organized as follows. The next subsections review types of foreign aid and the relevant literature on the links between foreign aid and human development. Section 2 describes the data, Section 3 looks at the empirical analysis, Section 4 discusses the results, Section 5 reviews limitations, and Section 6 concludes.

1.1 Foreign Aid

As the impact of foreign aid is contextualized, it is imperative to understand what it encompasses. Foreign aid broadly covers the exchange of money in the form of loans between countries, be those loans tied to the promise of structural change by recipients or otherwise.

Since the end of World War II, partnerships between high-income countries and lower-income countries via the provision of foreign aid have become more commonplace as have institutions that facilitate these exchanges.

These institutions lend medium and long-term capital for productive investment. For example, the World Bank offers financing through investment projects, development policy support, and results-based programs to reduce poverty, promote sustainable growth, and strengthen institutions. (Financing, 2024). All of these programs are financed partly by contributions from wealthier member states and partly by funds from capital markets. However, the extent to which these loans, often with rigorous repayment schemes, actually have a net benefit on the recipient countries is often called into question. This impact of foreign aid provision, specifically on gender equality (the UN's sustainable development goal 5), is the relationship that this paper intends to analyze.

1.2 Literature Review

It is essential to analyze the debate over foreign aid's effectiveness in stimulating development, which has demonstrated mixed results, specifically when considering evidence at both a micro and macro level. Indicators of economic and social performance often bear positive results when looking at aid interventions at the micro level, but these are not translated when analyzing the effects at the macro level. Examined evidence on aid effectiveness revealed that interventions appear to produce positive results when looking at individual targeted aid interventions, yet the impact is limited at the macroeconomic level (Ndikumana, 2012). The difficulty in observing trends of aid effectiveness at the macro level lies in that most aid targeted towards gender equality is spread across multiple dimensions such as health, education and rural development. Similarly, the country-level impact of foreign aid remains relatively untouched, as the donor community often analyses gender-targeted interventions only in relation to specific (micro) programs so they can pinpoint whether a program should specifically remain funded. Additionally, regardless of how gender performance is defined, the concept represents a complex set of dynamic processes at the macro level that are latent in certain observable indicators, which may or may not accurately reflect the underlying mechanisms.

Furthermore, the results that do exist within published work on aid effectiveness on gender performance indicate only a conditional effect at the macro level. Existing literature suggests mixed conclusions, implying aid has a significant impact dependent on initial human development and targeted implementation. It is, therefore, important to analyze aid effectiveness in the context that most aid targeted towards gender equality is implemented through gender mainstreaming; meaning gender is implemented through most development assistance as a secondary criterion but not as the primary goal.

For example, Bali Swain, Garikipati, and Wallentin suggest that aid alone, even when targeted to improve gender outcomes, does not show any shift in systematic inequalities (2020). They used a structural equation model with several traditional multivariate procedures, like factor analysis, regression analysis, discriminant analysis, and canonical correlation, to draw their results. The authors first used a factor analysis model to measure gender performance in

recipient countries from directly observable dimensions: maternal mortality rates, adolescent fertility rate, and proportion of seats held by women in parliament. Then, to understand the transferability of aid programmes, authors conducted a comparative analysis of factors to identify which were the most effective in having a significantly positive impact of foreign aid on gender performance in recipient countries. Given their empirical model, Bali Swain, Garikipati, and Wallentin then concluded that their results show that, at the macro level, foreign aid does not have a significant impact on improving the indicators used to measure gender performance in a country. Their reasoning proposes that results reflect the limited scale and quality of foreign aid targeted at women. Bali Swain, Garikipati, and Wallentin's work suggests that to prompt a successful shift in the gender status quo, foreign aid interventions need to be more directly targeted toward women's indicators of development (2022).

In parallel, Pickbourn and Ndikumana's study also pinpoints the conditionality of foreign aid effectiveness. They propose that the impact of aid on gender inequality is dependent on initial human development and per capita income (2016). In their work, authors use data on bilateral official aid disbursements at the project level from the OECD database with panel data techniques to control for country-specific effects using fixed-effects estimations, potential outliers using iterated reweighted least squares, and potential endogeneity of regressors using the generalized method of moments. The impact of foreign aid on gender inequality is then specifically investigated by estimating an econometric model where the UNDP's gender inequality index is regressed directly on total aid at the country level. Pickbourn and Ndikumana also use public expenditure on health and education, access to water, access to sanitation, the age dependency ratio and governance measured by a corruption index and a government stability index as control variables. Therefore, their results indicate the impact of foreign aid on overall gender inequality appears to depend on the initial level of human development and initial per capita income. Their conclusion is based on the observation that total aid reduces gender inequality in the absence of controls for initial conditions; yet when initial conditions are controlled for, the impact of total aid on gender inequality disappears. Pickbourn and Ndikumana's work, consequently, proposes that countries starting out with a higher level of human development or per capita income may be more successful in lowering gender inequality than countries with lower initial levels of human development and per capita income (2016). As a result, even with the existing literature, there is still a present gap in greater understanding of how foreign aid at the national level can impact gender equality.

2 - DATA

The dataset is a detailed collection of socio-economic, demographic, and aid-related variables from the World Bank Open data set, encompassing various countries over multiple years and designed for cross-country and over-time analysis. It covers 2002 to 2018, stopping before the COVID-19 pandemic to avoid outlying data caused by the global crisis.

To focus on developing countries and their gender equality outcomes, specific exclusions were made. Developed countries with strong gender equality and little reliance on foreign aid were removed as they were not relevant to the study and could incorrectly influence the result. Similarly, countries with negative foreign aid values were removed as it indicated their position as donors, which again mostly encompassed developed countries. Countries with limited or incomplete data were excluded to ensure reliable results, removing the ability to do random sampling and countries experiencing severe conflict or political instability, like South Sudan, Palestine (West Bank and Gaza), and Afghanistan, were excluded due to the uniqueness of their situations potentially distorting the findings. By narrowing the dataset to developing countries that rely on aid and show varying levels of gender equality, the analysis aims to provide clear and meaningful insights into how foreign aid affects gender outcomes in these contexts.

To construct our data set, the dependent variable used was an index based on principal component analysis (PCA) which includes the percentage of women in the labor force to indicate economic empowerment and gender equality, maternal mortality ratio to reflect gender equality in healthcare, and the percent of women in parliament to demonstrate gender participation in politics.

The net official development aid (ODA) received is used as an independent variable to measure each country's total foreign aid inflows. Net ODA includes concessional loans (loans with low interest rates and long repayment terms) and grants provided by official agencies of the Development Assistance Committee (DAC) members (a group of 32 major aid donors), multilateral institutions, and non-DAC countries. This aid is aimed at promoting economic development and improving welfare in countries listed as eligible for ODA assistance by the DAC.

The control variables encompass the following socio-economic and institutional variables: population, annual growth of GDP per individual, FDI as a percent of GDP, total school enrollment, government expenditure on education, and the rule of law to reflect governance quality and level of corruption, hence the political environment. These variables allow control over the fact that institutional differences may affect the impact of foreign aid on gender equality.

In addition to the primary dependent variables, some supplementary indicators were added to enrich the gender equality index. These indicators, popular in research and well-represented in the dataset, provide a broader view of gender equality. Female life expectancy at birth refers to the average number of years a newborn girl would expect to live under current mortality rates. This gives an idea of the general health conditions and health infrastructure for women. The adolescent fertility rate is commonly used and refers to the number of births per 1,000 women aged 15–19 years; it usually indicates early marriages and poor reproductive rights – both indicators in which gender inequality is a strong factor. The maternal mortality ratio, understood as the number of maternal deaths per 100,000 live births, is another critical health-related component of gender disparities.

Regional dummy variables were also incorporated to represent different geographical zones, accounting for geographical differences between regions that may influence gender equality or aid effectiveness. These variables facilitate comparisons across major geographical areas. However it is important to point out that fully representing certain regions is difficult – such as the Caribbean or the Middle East – and so there is a slight reduction of the precision for regional comparisons.

Time effects were represented by time dummy variables to account for global events or trends that uniformly affect all countries during given years. For example, the 2004 Indian Ocean Tsunami increased aid to the affected countries, such as Indonesia, and specific programs that supported women's recovery. Similarly, the 2010 Haiti Earthquake saw major increases in aid targeting women's health and safety. In this respect, the analysis controls for key global events that may impact aid flows and, subsequently, outcomes related to gender equality.

In sum, by focusing on developing countries that rely on foreign aid and exhibit varying gender equality outcomes, this dataset provides a robust framework to analyze the impact of aid on gender equality while accounting for socio-economic, institutional, and regional factors.

3 - EMPIRICAL ANALYSIS

3.1 Model Specification

The impact of foreign aid on gender inequality is analyzed using an econometric model with a linear probability approach based on a principal component analysis (PCA) index. The index combines key gender equality indicators including the percentage of women in the labor force, women in parliament, adolescent fertility rate, maternal mortality ratio, and female life expectancy. To ensure consistency in the direction of the variables, the adolescent fertility rate and maternal mortality ratio are inverted, as higher values represent worse outcomes, to align with the other indicators where higher values signify better gender equality. The PCA index, denoted as IQI, is used to rank countries each year. Then a dummy variable is created that equals 1 if a country's IQI is above the median for that year and 0 if it is below. By running a regression on this dummy variable, the coefficient of the independent variables can be interpreted as the probability of a country moving into the top half of gender equality rankings (1) for each additional unit increase in the explanatory variable or remaining in the bottom half (0) for a given year.

Furthermore, an important assumption in the model is that foreign aid does not immediately impact gender equality year-to-year as there is often a time lag before the effects of policy changes are realized within a country's government. To account for this delay, the independent variables in the model include time lagged values of foreign aid, ranging from $t-1$ to $t-3$, reflecting the time it takes for aid to influence gender equality outcomes.

The model will also incorporate control variables to account for a country's characteristics and economic conditions, including GDP per capita growth, population size, and political factors such as the "rule of law" variable, which is correlated with indicators like corruption and government effectiveness. Additionally, the model will consider the country's

position in international trade through foreign direct investment (FDI) as a percentage of GDP, and education, which will be weighted using two key variables. These control variables ensure that the independent variables are not confounded by other factors that could influence gender equality (such as those that may correlate with the dependent and independent variables) thus preventing potential bias and ensuring the model's accuracy by minimizing correlation with the error terms.

To minimize bias in the estimators of the independent variables, the model uses the characteristics of panel data to account for entity-specific fixed effects. In this study, these fixed effects represent unobserved variables that vary across countries but remain constant over time, such as cultural factors, which can strongly influence gender equality outcomes.

Additionally the model addresses time-specific effects: factors that vary over time but affect all countries similarly such as changes in international regulations or global shocks like climate events. These effects are captured using dummy variables and set to 1 or 0 for each year.

The model also considers geographical factors that may impact groups of countries in similar ways, such as the effects of global warming on certain continents or economic policies like those in the European Union, by creating continent-specific dummy variables.

Finally, the model assumes heteroskedasticity in the error terms, which is addressed using cluster estimation. This approach provides robust standard errors that account for the grouped structure of the data, ensuring more reliable inference. The regression model is thus defined in the following equation:

$$Y_{i,t} = \beta X' + \theta Z' + \alpha_i + \sum_{s=1}^{T-1} \delta_s \cdot 1 \cdot (t = s) + \sum_{r=1}^{R-1} \gamma_r \cdot 1 \cdot (r = i) + v_{i,t} \quad (1)$$

X' is the transpose of a 4x1 vector of the independent variables used: the lagged values of foreign aid from t-1 up to t-3 and the value 1 for β_0 that is included in the β 4x1 vector representing the coefficient of the explanatory variables. Similarly, Z' is the transpose of a mx1 vector including the m control variables that the model decides to use on a particular analysis and with θ once again a mx1 vector of coefficients. The α_i represents the entity specific fixed effects that remain constant through time. Through the expression $\sum_{s=1}^{T-1} \delta_s \cdot 1 \cdot (t = s)$, the model accounts for the time specific effects that remain constant across entities with δ_s representing a dummy variable set equal to 1 if the model looks at the year of interest in a particular iteration. Finally, the expression $\sum_{r=1}^{R-1} \gamma_r \cdot 1 \cdot (r = i)$ looks at the regional effect with γ_r being a dummy variable for the different continents while $v_{i,t}$ denotes the error term.

Before estimating the causal effect through the regression model, it is important to look at some summary statistics of the key variables described in the following tables. The relationships between these variables provide an intuitive basis for arguing that, by including the control variables, we satisfy the critical assumption $E[u|X] = 0$ and ensure that the error term is uncorrelated with the explanatory variables.

Table 1. Summary Statistics of Dependent and Independent variables for All Countries

Summary Statistics for All Countries					
Variable	Observations	Mean	Std. Dev.	Min	Max
Net official development received	1963	593000000.0	974000000.0	-939000000.0	2210000000.0
IQI	1946	7.19E-10	1.000002	-4.431685	0.9187564
Percent of women in parliament	1946	16.65154	10.80526	0.0	63.75
Adolescent fertility rate	2030	72.60966	44.08411	4.817	205.221
Maternal mortality ratio	2030	259.2389	284.9154	1.0	1668.0
Percent of women in labor force	2030	40.50301	8.874509	12.15103	53.71009
Female life expectancy at birth	2030	68.7139	8.968076	43.356	83.149

Table 2. Summary Statistics of Dependent and Independent variables for Countries in Europe

Variable	Observations	Mean	Std. Dev.	Min	Max
Net Official Development Aid Received	167	353000000.0	324000000.0	-250000000.0	2080000000.0
IQI	162	0.8557084	0.0504512	0.7231868	0.9187564
Upper Half	162	1.0	0.0	1.0	1.0
Percent of Women in Parliament	162	18.52269	8.820085	3.053435	38.33333
Adolescent Fertility Rate	171	26.34623	12.8428	10.319	64.142
Maternal Mortality Ratio	171	17.78947	13.38226	1.0	54.0
Percent of Women in Labor Force	171	45.75041	4.780319	32.1404	53.57444
Female Life Expectancy at Birth	171	76.81861	2.512208	69.884	81.608

Table 3. Summary Statistics of Dependent and Independent variables for Countries in Africa

Variable	Observations	Mean	Std. Dev.	Min	Max
Net Official Development Aid Received	845	749000000.0	893000000.0	-14500000.0	11400000000.0
IQI	810	-0.7984974	1.051296	-4.431685	0.78223
Upper Half	810	0.1111111	0.3144639	0.0	1.0
Percent of Women in Parliament	810	17.54886	11.27604	0.0	63.75
Adolescent Fertility Rate	857	102.5236	45.7956	4.817	205.221
Maternal Mortality Ratio	857	481.9137	302.4635	38.0	1668.0
Percent of Women in Labor Force	857	42.92761	7.923019	14.35356	53.71009
Female Life Expectancy at Birth	857	60.99527	7.657604	43.356	80.694

Table 4. Summary Statistics of Control Variables for All Countries

Variable	Observations	Mean	Std. Dev.	Min	Max
FDI as percent of GDP	2030	4.5	2.2	0.1	15.0
Total school enrollment	2030	85.6	10.8	50.0	100.0
GDP per capita growth	2030	2.1	1.5	-3.0	10.0
Rule of law	2030	-0.35	0.7	-1.5	1.5
Percent of government expenditure for education	2030	4.8	1.2	2.3	8.5
Population	2030	34000000.0	51000000.0	100000.0	140000000.0

Table 5. Summary Statistics of Control Variables for Countries in Europe

Variable	Observations	Mean	Std. Dev.	Min	Max
FDI as percent of GDP	161	7.39025	7.823535	0.5358079	55.07288
Total school enrollment	126	92.06448	8.745918	74.53892	116.7629
GDP per capita growth	171	0.1115413	0.1557977	-0.3029255	0.5667818
Rule of law	171	-0.4580168	0.329798	-1.303661	0.3357719
Percent of government expenditure for education	116	3.929213	1.553124	1.86034	9.50976
Population	171	4929204.0	6559617.0	609828.0	81400000.0

Table 6. Summary Statistics of Control Variables for Countries in Africa

Variable	Observations	Mean	Std. Dev.	Min	Max
FDI as percent of GDP	839	4.470296	7.98444	-10.03838	103.3374
Total school enrollment	507	45.49644	23.68972	6.07717	111.6274
GDP per capita growth	850	0.0710704	0.1610736	-0.5135109	1.16661
Rule of law	848	-0.7120536	0.6297528	-2.590877	1.023956
Percent of government expenditure for education	587	4.070336	2.111494	0.3930456	13.21957
Population	857	19100000.0	27600000.0	149841.0	198000000.0

Table 7. Correlation between all variables for all countries

Variable	netofficialdevelopmentaidreceive	IQI	top half	adolescentfertilityrate	percentofwomenlaborforce	fdiaspercentofgdp	totalschoolenrollment	rule_of_law	gov_expenditure_on_education
netofficialdevelopmentaidreceive	1.0	-0.2733	-0.4276	0.2077	-0.1734	-0.1084	-0.3618	-0.2095	-0.2066
IQI	-0.2733	1.0	0.728	-0.6991	-0.2267	0.1378	0.7651	0.4243	0.1856
top half	-0.4276	0.728	1.0	-0.5917	-0.1126	0.1056	0.7353	0.4058	0.1008
adolescentfertilityrate	0.2077	-0.6991	-0.5917	1.0	0.2303	0.0452	-0.7138	-0.3623	-0.1909
percentofwomenlaborforce	-0.1734	-0.2267	-0.1126	0.2303	1.0	0.1627	0.1082	-0.1022	0.1003
fdiaspercentofgdp	-0.1084	0.1378	0.1056	0.0452	0.1627	1.0	0.1082	0.1242	0.1003
totalschoolenrollment	-0.3618	0.7651	0.7353	-0.7138	0.1082	0.1082	1.0	0.418	0.232
rule_of_law	-0.2095	0.4243	0.4058	-0.3623	-0.1022	0.1242	0.418	1.0	0.2128
gov_expenditure_on_education	-0.2066	0.1856	0.1008	-0.1909	0.1003	0.1003	0.232	0.2128	1.0

Tables 1, 2, and 3 present the summary statistics for official development aid, our gender equality index (IQI), the dummy variable for rank, and the five gender equality indicators used in our analysis. Here, we can observe key patterns both at the aggregate level and when comparing the European and African continents. From Table 1, the aggregate mean of the IQI is approximately 0 (7.19×10^{-10}), with values ranging from -4.5 to 1. However, when we break it down by region, the mean IQI for Europe is much higher at 0.86, close to the maximum value of 1; meanwhile for Africa it is significantly lower at -0.86. European countries also consistently outperform African countries across all gender equality indicators. For example, the mean value of the "upper half" dummy variable is 1 for Europe, indicating that all European countries rank in the top half for gender equality. In contrast, only 10% of African countries fall into this category.

One notable feature of our data is the disparity in regional representation: Europe is represented by about 10 countries (with 170 observations, or 17 per country), while Africa is represented by 50 out of its 54 countries, making the African sample much more representative of the continent as a whole. Regarding foreign aid, African countries in our sample receive an

average of \$750 million annually compared to \$350 million for European countries. Given the differences in gender equality indicators and development levels (discussed further in Tables 4, 5, and 6) on control variables, the higher aid levels for Africa are expected and form a key focus of our regression analysis.

Turning to the control variables, we first examine the aggregate means across our sample countries. We compare Europe and Africa with the goal of understanding the relationships between variables, identifying potential causality, and addressing omitted variable bias in our regression model. For example, the average FDI as a percentage of GDP is higher for European countries (over 7%) compared to the African average of 4.47%, suggesting that European countries are more integrated into the global economy. This variable is included as a control to account for the economic role of countries in global trade.

We also observe significant differences in two key control variables – total school enrollment and the rule of law – between continents which helps to explain variations in gender equality and official aid. Specifically, the mean rule of law score for European countries is -0.45, while for African countries it is below -0.7. Similarly, European countries have an average school enrollment rate of over 90%, compared to just 45% in Africa. These differences support the inclusion of these variables in our analysis, as they help mitigate omitted variable bias. Based on existing research in economic development, we argue that education and the rule of law are critical factors in explaining both gender equality outcomes and the distribution of aid, as further supported by the correlations in Table 7. However, it is important to note that these relationships do not imply causality, a point we address in our regression model.

As expected, the differences observed between African and European countries in gender equality, education, and political conditions are indicated by strong positive correlations. For instance, the correlation between the IQI and total school enrollment is 0.77, and the correlation between the IQI and rule of law exceeds 0.4. Given that poor education or weak political conditions can influence the amount of aid received, including these variables in our analysis is essential to reduce omitted variable bias and ensure the validity of our regression model's assumptions.

Finally, despite observing a negative correlation between IQI and net official development aid in Table 7, we refrain from concluding that foreign aid negatively impacts gender equality. We hypothesize that aid may require time to show measurable effects, and this will be explored further in our regression analysis.

4 - ECONOMETRIC RESULTS

This section presents the main regression results on the impact of foreign aid on gender equality in recipient countries, showing that it has little to no significant positive effect on gender indicators. We will first present the results without control variables, followed by a stepwise addition of control variables in each subsequent table. Table 8 displays the regression results for overall gender inequality, measured by the Gender Inequality Index (IQI), without accounting for any control variables.

Table 8: Regression on Gender Equality on Foreign Aid without Control Variables

Variable	Coefficient	Std. Error	t-Statistic	p-Value
Lagged Aid (t-1)	6.4E-12	1.7E-12	3.77	0.0
Lagged Aid (t-2)	3.29E-12	1.45E-12	2.27	0.025
Lagged Aid (t-3)	4.95E-12	2.22E-12	2.23	0.028
Continent: Asia	0.0118106	0.0137696	0.86	0.393
Continent: Africa	Omitted			
Continent: Oceania	Omitted			
Continent: South America	Omitted			
Continent: North America	Omitted			
Year: 2006	-0.0102518	0.0091242	-1.12	0.263
Year: 2007	-0.0197506	0.0128417	-1.54	0.127
Year: 2008	-0.0113267	0.0153956	-0.71	0.479
Year: 2009	-0.0199163	0.0129047	-1.54	0.125
Year: 2010	-0.0221039	0.0180329	-1.23	0.219
Year: 2011	0.0026511	0.0179068	-0.15	0.883
Year: 2012	-0.0111848	0.0204263	-0.56	0.560
Year: 2013	-0.0231128	0.0242263	-0.95	0.341
Year: 2014	-0.0307644	0.0203977	-1.47	0.142
Year: 2015	-0.0213568	0.0202397	-1.06	0.290
Year: 2016	-0.0125658	0.0223307	-0.56	0.579
Year: 2017	-0.0121512	0.0222338	-0.55	0.581
Year: 2018	-0.0215731	0.0223338	-0.97	0.336
Constant	0.4861956	0.0172524	28.18	0.000
R-squared: Overall	0.0155			

Analyzing the indicators in Table 8, we find that the lagged aid coefficients are positive, suggesting that each additional unit of foreign aid increases the likelihood of a country moving from the lower half to the upper half of the gender equality ranking (from 0 to 1) by the amount specified in the coefficient. The p-values for these coefficients are significant at the 5% level, indicating that additional aid does indeed contribute to improving gender equality. However, the small magnitude of these coefficients is expected, as they represent the effect of a single unit of aid, whereas foreign aid is typically distributed in much larger amounts, often in the millions or billions.

In addition, we find that the R-squared value was relatively strong and increased as control variables were added, confirming their expected effect. Notably, while some continents were omitted from the analysis, most countries remained unchanged between states 1 and 0. The only continent showing significant changes was Asia, with possible changes in Europe, though Europe was not included in this analysis. This suggests that being in Asia increases the likelihood of moving to the upper half of the rank, as indicated by the positive coefficient for the dummy variable. However, given the small sample size, the external validity of this result is limited, and generalization to a broader population is not possible.

Additionally, including control variables led to a significant shift in the coefficients for lagged aid, further supporting the increase in R-squared and confirming that these variables improved our regression model. Without them, such effects would not have been identifiable based solely on the R-squared value.

Lastly, another key finding is that, over time, the probability of a country moving to the upper half of the rank decreases. While causality cannot be inferred, the negative coefficient for time suggests that the effectiveness of aid may have diminished over the years, which warrants further investigation.

The regression results for overall gender inequality, as captured in the gender inequality index (IQI) and accounting for GDP per capita growth and population control variables, are presented in Table 9.

Table 9: Regression of Gender Equality on Foreign Aid with GDP per capita growth and population Control Variables

Variable	Coefficient	Std. Error	t-Statistic	p-Value
Lagged Aid (t-1)	6.68E-12	1.77E-12	3.78	0.000
Lagged Aid (t-2)	3.41E-12	1.49E-12	2.30	0.023
Lagged Aid (t-3)	5.05E-12	2.25E-12	2.25	0.026
Lagged GDP per Capita	-0.0271327	0.025724	-1.05	0.294
Population	2.67E-11	2.67E-11	0.10	0.921
Continent: Africa (omitted)	Omitted			
Continent: Asia	0.0147807	0.0131072	1.13	0.262
Continent: Oceania (omitted)	Omitted			
Continent: South America (omitted)	Omitted			
Continent: North America (omitted)	Omitted			
Year: 2006	-0.0107218	0.0092042	-1.16	0.246
Year: 2007	-0.0198226	0.0129847	-1.53	0.130
Year: 2008	-0.0113308	0.0162136	-0.70	0.486
Year: 2009	-0.0195316	0.0130638	-1.50	0.138
Year: 2010	-0.0245249	0.0219389	-1.12	0.266
Year: 2011	-0.0039215	0.0189503	-0.21	0.836
Year: 2012	-0.0128741	0.0210649	-0.60	0.552
Year: 2013	-0.0245567	0.024461	-1.00	0.320
Year: 2014	-0.033727	0.0233465	-1.44	0.151
Year: 2015	-0.0167163	0.0235325	-0.71	0.479
Year: 2016	-0.0279281	0.0272332	-1.03	0.307
Year: 2017	-0.0172757	0.0266997	-0.65	0.519
Year: 2018	-0.0244758	0.02536061	-0.96	0.337
Constant	0.4887657	0.0130631	37.42	0.000
R-squared: Overall	0.0187			

Examining the indicators in Table 9, we see that the lagged aid coefficients remain positive and grow in magnitude. The p-value also remains significant at the 5% significance level, suggesting that additional aid further contributes to improved gender equality, even more so now that some control variables are considered.

That said, we observe an increase in the R-squared value with the inclusion of GDP per capita growth and population as control variables, which is expected given the addition of more variables.

Moreover, it is important to note the p-value for the population is far from significant at the 5% level, suggesting that population size has no meaningful impact on gender equality and may warrant further investigation. Similarly, the p-value for GDP per capita growth is also not significant at the 5% level, indicating that it is not a strong explanatory variable for gender equality. Nevertheless, their inclusion slightly refines our model as evidenced by minor changes in the estimators for lagged aid. Furthermore, the regression results for overall gender inequality, as captured in the gender inequality index (IQI), with FDI as a Percentage of GDP included as a control variable are presented in Table 10,

Table 10: Regression of Gender Equality on Foreign Aid with FDI as a Percentage of GDP Added as a Control Variable

Variable	Coefficient	Std. Error	t-Statistic	p-Value
Lagged Aid (t-1)	6.7E-12	1.78E-12	3.77	0.0
Lagged Aid (t-2)	3.49E-12	1.49E-12	2.34	0.021
Lagged Aid (t-3)	5.11E-12	2.24E-12	2.28	0.024
Lagged GDP per Capita	-0.0272837	0.0260645	-1.05	0.297
Population	3.18E-11	2.72E-11	0.12	0.907
FDI as % of GDP	-0.0003664	0.002586	-1.42	0.159
Continent: Africa (omitted)	Omitted			
Continent: Asia	0.0145673	0.0132116	1.1	0.272
Continent: Oceania (omitted)	Omitted			
Continent: South America (omitted)	Omitted			
Continent: North America (omitted)	Omitted			
Year: 2006	-0.0101046	0.0090603	-1.12	0.267
Year: 2007	-0.0191915	0.0127384	-1.51	0.135
Year: 2008	-0.010621	0.0160716	-0.66	0.51
Year: 2009	-0.0193059	0.0129956	-1.49	0.14
Year: 2010	-0.0239733	0.021819	-1.1	0.276
Year: 2011	-0.003172	0.0188686	-0.17	0.867
Year: 2012	-0.0122572	0.0215093	-0.57	0.57
Year: 2013	-0.0242222	0.0246009	-0.98	0.327
Year: 2014	-0.033768	0.0234511	-1.44	0.152
Year: 2015	-0.0164442	0.0235581	-0.7	0.487
Year: 2016	-0.0218942	0.0274212	-1.03	0.306
Year: 2017	-0.0172457	0.0268403	-0.64	0.522
Year: 2018	-0.0247444	0.0255772	-0.97	0.335
Constant	0.4876374	0.0133083	36.64	0.0
R-squared: Overall	0.0187			

Examining the indicators in Table 10, we see that the lagged aid coefficients remain positive, with p-values significant at the 5% level. This reinforces the finding that lagged foreign aid continues to have an effect on gender inequality even after including FDI in the model.

Interestingly, the coefficient for FDI is negative, which is unexpected given that we would intuitively assume higher FDI reflects a positive economic image internationally, potentially enhancing gender equality. However, the negative result suggests a different dynamic. One possible explanation is that countries with greater access to FDI may prioritize these investment opportunities over foreign aid as FDI typically comes with fewer requirements for infrastructure changes, unlike the more conditional foreign aid. Furthermore, the regression results for overall gender inequality with all control variables, as captured in the gender inequality index (IQI), are presented in Table 11.

Table 11: Regression of Gender Equality on Foreign Aid with all Control Variables

Variable	Coefficient	Std. Error	t-Statistic	p-Value
Lagged Aid (t-1)	1.39E-11	1.1E-11	1.26	0.21
Lagged Aid (t-2)	-5.21E-12	6.18E-12	-0.84	0.401
Lagged Aid (t-3)	1.09E-11	1E-11	1.08	0.282
Lagged GDP per Capita	-0.0237539	0.0386931	-0.61	0.541
Population	3.46E-11	3.07E-10	0.11	0.911
FDI as % of GDP	-0.0010177	0.0007213	-1.41	0.161
Total School Enrollment	0.002103	0.0015401	1.37	0.175
Gov. Expenditure on Education	-0.0062661	0.0062508	-1.0	0.319
Rule of Law	-0.023095	0.043842	-0.53	0.599
Continent: Africa (omitted)	Omitted			
Continent: Asia	0.0290441	0.0211309	1.37	0.172
Continent: Oceania (omitted)	Omitted			
Continent: South America (omitted)	Omitted			
Continent: North America (omitted)	Omitted			
Year: 2006	0.0038656	0.0303233	0.13	0.899
Year: 2007	0.0092851	0.0310874	0.3	0.766
Year: 2008	-0.0088987	0.0346578	-0.26	0.798
Year: 2009	0.0467568	0.0245325	1.91	0.059
Year: 2010	-0.0273736	0.0305435	-0.9	0.372
Year: 2011	-0.0115468	0.0237174	-0.49	0.627
Year: 2012	0.0376402	0.0272179	1.38	0.17
Year: 2013	0.0382531	0.0291342	1.31	0.192
Year: 2014	-0.0590867	0.0367205	-1.61	0.111
Year: 2015	0.0467501	0.0281982	1.66	0.1
Year: 2016	-0.0665648	0.0364452	-1.83	0.071
Year: 2017	-0.0593256	0.040473	-1.47	0.146
Year: 2018	-0.059623	0.034247	-1.74	0.085
Constant	0.3977983	0.1201073	3.31	0.001
R-squared: Overall	0.2858			

Observing the specific indicators in Table 11, we note a significant shift in the lagged aid coefficients. Specifically, Lagged Aid (t-2) became negative, while Lagged Aid (t-1) and Lagged Aid (t-3) increased in magnitude. Moreover, their p-values are no longer significant at the 5% level, indicating a change from previous results and suggesting the presence of omitted variable bias in earlier models.

Including two key control variables, Total School Enrollment and Government Expenditure on Education, appears to have addressed this bias, as evidenced by the notable shift in the significance of the lagged aid coefficients. This finding underscores the importance of education as a critical factor influencing gender equality and validates its role as an essential control variable.

In addition, introducing the Rule of Law as a control variable further enhances the model. Given that the rule of law often correlates with other political factors, such as corruption, its inclusion strengthens the explanatory power of the model. This is reflected in the substantial 40% increase in the R-squared value, which rises to 0.2858, compared to lower values in previous regressions. This sharp increase suggests that the added control variables, education and rule of law, are key factors in explaining gender inequality and significantly improve the model's fit.

Looking at the specific results in the coefficients of the added control variables, we can see the rule of law has a negative coefficient, going against the intuition that weaker political and legal systems are associated with lower gender equality – this requires further investigation.

Furthermore, Total school enrollment shows a positive coefficient, which aligns with our hypothesis, although the significance level is unimportant at 17%. However, the coefficient for expenditure on education is unexpectedly negative, despite not being statistically significant at the 5% level. This negative sign is surprising, as we would typically expect education spending to have a positive effect on gender equality. A potential explanation for this negative correlation is that countries with lower school enrollment rates may be spending more on education in an attempt to address these gaps, rather than broadly improving access, which could explain the negative relationship between spending and enrollment. Therefore, the inclusion of education and rule of law as control variables has notably improved the model, addressing prior omitted variable bias and enhancing its explanatory power. However, the unexpected findings, such as the negative relationship between education spending and gender equality, highlight the need for further investigation to better understand the underlying dynamics and refine our understanding of how these factors interact.

5 - LIMITATIONS

While the model provides accurate estimates of the effect of foreign aid on developing countries, it is important to recognize the limitations inherent in both the data collection process and the methodology. Understanding these limitations is crucial for assessing the validity of the model and informing future improvements. Before delving into specific limitations, however, it is important to note that time constraints, particularly as students, represent a significant

challenge in the analysis. With this in mind, this section will explore the limitations of the data, the model, and the variables used in the study.

5.1 Limitations of the Data

One key challenge encountered during the initial modeling stages was the nature of the gender equality indicators used as dependent variables. Variables like female life expectancy and adolescent fertility rates change only incrementally each year, leading to multicollinearity within the data. To address this, we initially extended the time intervals, shifting from annual to four-year periods, under the assumption that larger time gaps would allow for more meaningful changes. However, even with these longer intervals, the changes were still too small to fully resolve the collinearity issue. As a result, we used a linear probability model, which is less sensitive to multicollinearity, to proceed with the analysis.

Another limitation stemmed from gaps in data availability. Countries such as Haiti and Iraq lacked data on key gender equality indicators and control variables like total school enrollment, limiting our sampling method and preventing us from using random sampling, which would have ensured more representative data. As a result, some countries were excluded, reducing the scope of our analysis and limiting our ability to include additional control variables.

Additionally, the dataset for net foreign aid received contained some negative values, suggesting that certain countries received less aid than they distributed. For example, Gabon showed a negative value, which seemed illogical given its status as a net aid recipient. Despite reviewing the data documentation, no explanation was found for these anomalies. One possible explanation is that these negative values could represent repayments of previous loans, but without confirmation from the data source, this uncertainty may have introduced bias into the model.

Finally, while the "rule of law" was included as a control variable to capture political stability, it showed limited significance in the model. This may be because "rule of law" was not as strongly correlated with other political stability indicators, like corruption, as expected. A different control variable, such as corruption levels, might have provided a better measure of political stability and improved the model's robustness.

5.2 Limitations of the Regression Model

In planning the regression model, the Arellano-Bond estimator was initially considered for its strength in panel data analysis. However, due to multicollinearity issues, the estimator proved unsuitable as it requires no autocorrelation in the errors (Stata Press, 2023). While a more advanced panel data regression method could have improved the model, we were unable to identify a suitable one and this limitation persisted.

Additionally, the model assumes that all foreign aid equally contributes to gender equality, which overlooks the fact that different types of aid may have varying impacts. For instance, aid tied to democratic institution-building may not have the same effect on gender equality as more general aid, and the current model cannot account for this distinction.

Finally, while a secondary regression on the gender equality rank was initially planned to provide further insights, time constraints and the already extensive analysis led to its exclusion.

Including this additional regression could enhance the variety of results and offer a deeper understanding of the impact.

5.3 Dummy Variables

Including multiple dummy variables in the model strengthens the analysis and improves result reliability, but there are some limitations amongst those included. For example, the regional dummy variables could have been further divided into more specific regions, such as the Middle East and Latin America, to better reflect cultural differences. However, breaking down regions into smaller areas could have led to too few countries in each group, making the variables less significant and less useful for model refinement. As a result, a more generalized regional approach was adopted.

Additionally, Europe was used as the β_0 variable, which is necessary for constructing the regional dummies, but this means the impact of being in Europe was not directly accounted for in the model. With more time, this limitation could have been addressed by rerunning the regression with Oceania as the β_0 variable, allowing for a more meaningful comparison of European countries.

Lastly, the time dummy variables present a limitation because the first three years (2002–2004) are excluded from the analysis. This is due to the three-year lag period used in the model, which means the earliest available data for those years could not be incorporated, reducing the analysis' scope for that time frame.

6 - CONCLUSION

The findings of this study contribute valuable insights to the ongoing understanding of foreign aid in improving gender equality in developing nations. Despite the increase in the global push for gender equality, our results highlight that once all controlled variables are accounted for, the positive macro-level impact of foreign aid on gender equality remains limited on initial conditions in developing nations.

The findings emphasize that, while foreign aid can contribute to progress on gender equality, its impact is neither uniform nor transformative without supportive domestic policies and robust institutions. The mixed results align with existing literature, which often finds that aid effectiveness is contingent on factors such as initial human development levels, political stability, and aid targeting gender-specific goals.

These results also underline the importance of integrating gender-sensitive approaches into development aid strategies to enhance their effectiveness. The study sheds light on the conditional nature of aid effectiveness and calls for the rethinking of aid strategies to ensure the increase of gender equality is not only included in foreign aid provision, but becomes a main priority goal at an international level.

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APPENDIX

Table of Countries Used in Empirical Analysis

Continent	Country
Africa	Algeria, Angola, Benin, Botswana, Burkina Faso, Burundi, Cabo Verde, Cameroon, Central African Republic, Chad, Comoros, Congo, Dem. Rep., Congo, Rep., Côte d'Ivoire, Djibouti, Equatorial Guinea, Eritrea, Eswatini, Ethiopia, Gabon, Gambia, The, Ghana, Guinea-Bissau, Kenya, Lesotho, Liberia, Libya, Madagascar, Malawi, Mali, Mauritania, Mauritius, Morocco, Mozambique, Namibia, Niger, Nigeria, Rwanda, Sao Tome and Principe, Senegal, Sierra Leone, Somalia, South Africa, South Sudan, Sudan, Tanzania, Togo, Tunisia, Uganda, Zambia, Zimbabwe
Asia	Bahrain, Bangladesh, Bhutan, Cambodia, India, Indonesia, Iraq, Jordan, Kazakhstan, Kyrgyz Republic, Lao PDR, Lebanon, Malaysia, Maldives, Mongolia, Myanmar, Nepal, Oman, Pakistan, Philippines, Saudi Arabia, Sri Lanka, Tajikistan, Thailand, Timor-Leste, Turkiye, Turkmenistan, Uzbekistan
North America	Barbados, Belize, Costa Rica, Dominican Republic, El Salvador, Guatemala, Haiti, Honduras, Jamaica, Mexico, Nicaragua, Panama, Trinidad and Tobago
Oceania	Papua New Guinea, Samoa, Solomon Islands, St. Lucia, St. Vincent and the Grenadines, Tonga, Vanuatu
South America	Argentina, Bolivia, Brazil, Chile, Colombia, Ecuador, Guyana, Paraguay, Peru, Suriname, Uruguay

Selected Countries Results

Bosnia and Herzegovina

Regression results for Bosnia and Herzegovina:

OLS Regression Results

Dep. Variable:	lqi	R-squared:	0.422
Model:	OLS	Adj. R-squared:	-0.011
Method:	Least Squares	F-statistic:	0.4028
Date:	Thu, 28 Nov 2024	Prob (F-statistic):	0.802
Time:	15:40:47	Log-Likelihood:	31.165
No. Observations:	15	AIC:	-48.33
Df Residuals:	8	BIC:	-43.37
Df Model:	6		
Covariance Type:	HC3		

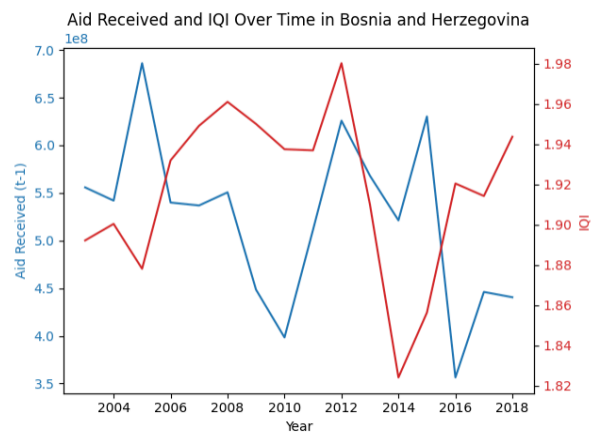
	coef	std err	z	P> z	[0.025	0.975]
const	1.7657	0.237	7.443	0.000	1.301	2.231
Net Official Development Aid Received t-1	-1.162e-10	2.27e-10	-0.511	0.609	-5.62e-10	3.3e-10
Net Official Development Aid Received t-2	-1.224e-10	1.66e-10	-0.736	0.462	-4.48e-10	2.03e-10
growth_lqi	0.0097	0.010	0.938	0.348	-0.011	0.030
lagged_GDPperCAPITA	0.0107	0.142	0.132	0.895	-0.259	0.297
FDI as Percent of GDP	-0.0034	0.009	-0.403	0.687	-0.020	0.013
Population	7.692e-08	7.48e-08	1.028	0.304	-6.97e-08	2.24e-07
Continent_Africa	0	0	nan	nan	0	0
Continent_Asia	0	0	nan	nan	0	0
Continent_North America	0	0	nan	nan	0	0
Continent_Oceania	0	0	nan	nan	0	0
Continent_South America	0	0	nan	nan	0	0

Omnibus:	0.569	Durbin-Watson:	1.050
Prob(Omnibus):	0.753	Jarque-Berra (JB):	0.588
Skew:	0.156	Prob(JB):	0.745
Kurtosis:	2.082	Cond. No.	inf

Notes:

[1] Standard Errors are heteroscedasticity robust (HC3)

[2] The smallest eigenvalue is 0. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.



Morocco

Regression results for Morocco:

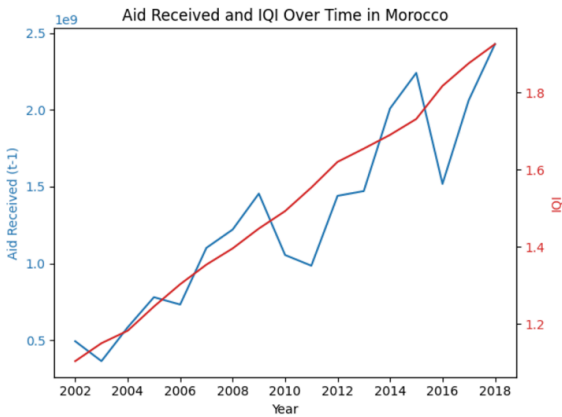
OLS Regression Results

Dep. Variable:	iqi	R-squared:	0.998
Model:	OLS	Adj. R-squared:	0.997
Method:	Least Squares	F-statistic:	0.4254
Date:	Thu, 28 Nov 2024	Prob (F-statistic):	0.740
Time:	15:40:47	Log-Likelihood:	51.818
No. Observations:	16	AIC:	-89.64
Df Residuals:	9	BIC:	-84.23
Df Model:	6		
Covariance Type:	HC3		

	coef	std err	z	P> z	[0.025	0.975]
Net Official Development Aid Received t-1	-9.884e-12	6.29e-11	-0.157	0.875	-1.33e-10	1.13e-10
Net Official Development Aid Received t-2	-1.706e-11	2.86e-11	-0.596	0.551	-7.32e-11	3.9e-11
growth_iqi	0.0070	0.007	0.962	0.336	-0.007	0.021
lagged_GDPperCAPITA	0.0225	0.232	0.097	0.923	-0.433	0.478
FDI as Percent of GDP	0.0005	0.008	0.056	0.956	-0.018	0.019
Population	1.308e-07	1.81e-08	7.233	0.000	9.53e-08	1.66e-07
Continent_Africa	-2.7472	0.546	-5.035	0.000	-3.817	-1.678
Continent_Asia	0	0	nan	nan	0	0
Continent_North America	0	0	nan	nan	0	0
Continent_Oceania	0	0	nan	nan	0	0
Continent_South America	0	0	nan	nan	0	0

Omnibus: 1.766 Durbin-Watson: 0.004
Prob(Omnibus): 0.414 Jarque-Bera (JB): 1.157
Skew: -0.388 Prob(JB): 0.561
Kurtosis: 1.935 Cond. No. inf

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
[2] The smallest eigenvalue is 0. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.



Mauritius

Regression results for Mauritius:

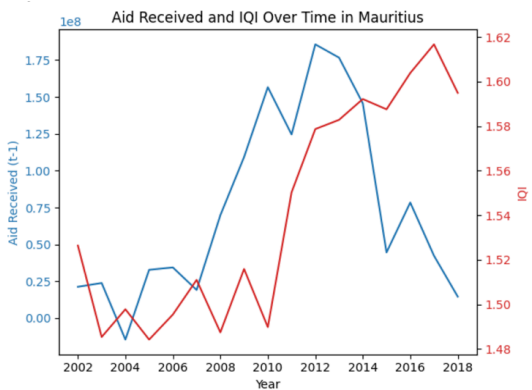
OLS Regression Results

Dep. Variable:	iqi	R-squared:	0.811
Model:	OLS	Adj. R-squared:	0.686
Method:	Least Squares	F-statistic:	2.351
Date:	Thu, 28 Nov 2024	Prob (F-statistic):	0.132
Time:	15:40:47	Log-Likelihood:	35.007
No. Observations:	16	AIC:	-64.01
Df Residuals:	9	BIC:	-58.61
Df Model:	6		
Covariance Type:	HC3		

	coef	std err	z	P> z	[0.025	0.975]
Net Official Development Aid Received t-1	-1.006e-10	2.59e-10	-0.698	0.485	-6.88e-10	3.27e-10
Net Official Development Aid Received t-2	1.646e-10	3.22e-10	0.511	0.609	-4.66e-10	7.96e-10
growth_iqi	0.0032	0.008	0.397	0.691	-0.013	0.019
lagged_GDPperCAPITA	-0.0091	0.158	-0.061	0.952	-0.303	0.285
FDI as Percent of GDP	-0.0071	0.009	-0.815	0.415	-0.024	0.010
Population	2.823e-06	1.2e-06	2.351	0.019	4.69e-07	5.18e-06
Continent_Africa	-1.9602	1.477	-1.327	0.184	-4.854	0.934
Continent_Asia	0	0	nan	nan	0	0
Continent_North America	0	0	nan	nan	0	0
Continent_Oceania	0	0	nan	nan	0	0
Continent_South America	0	0	nan	nan	0	0

Omnibus: 0.933 Durbin-Watson: 1.271
Prob(Omnibus): 0.627 Jarque-Bera (JB): 0.768
Skew: 0.271 Prob(JB): 0.674
Kurtosis: 2.057 Cond. No. inf

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
[2] The smallest eigenvalue is 0. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.



Brazil

Regression results for Brazil:

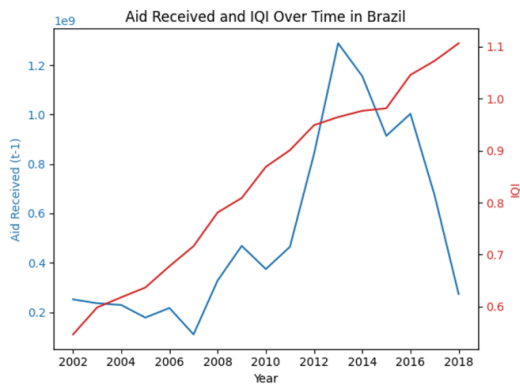
OLS Regression Results

Dep. Variable:	iqi	R-squared:	0.994
Model:	OLS	Adj. R-squared:	0.998
Method:	Least Squares	F-statistic:	0.7821
Date:	Thu, 28 Nov 2024	Prob (F-statistic):	0.533
Time:	15:40:47	Log-Likelihood:	47.258
No. Observations:	16	AIC:	-80.52
Df Residuals:	9	BIC:	-75.11
Df Model:	6		
Covariance Type:	HC3		

	coef	std err	z	P> z	[0.025	0.975]
Net Official Development Aid Received t-1	4.329e-11	1.91e-11	2.260	0.023	5.89e-12	8.07e-11
Net Official Development Aid Received t-2	-5.721e-11	5e-11	-1.145	0.252	-1.55e-10	4.07e-11
growth_iqi	0.0015	0.003	0.487	0.626	-0.004	0.007
lagged_GDPperCAPITA	-0.0224	0.071	-0.316	0.752	-0.161	0.116
FDI as Percent of GDP	0.0180	0.009	1.120	0.263	-0.007	0.027
Population	1.939e-08	1.51e-09	12.821	0.000	1.64e-08	2.24e-08
Continent_Africa	0	0	nan	nan	0	0
Continent_Asia	0	0	nan	nan	0	0
Continent_North America	0	0	nan	nan	0	0
Continent_Oceania	0	0	nan	nan	0	0
Continent_South America	-2.9911	0.264	-11.334	0.000	-3.508	-2.474

Omnibus: 2.517 Durbin-Watson: 1.110
Prob(Omnibus): 0.284 Jarque-Bera (JB): 1.420
Skew: -0.462 Prob(JB): 0.492
Kurtosis: 1.840 Cond. No. inf

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
[2] The smallest eigenvalue is 0. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.



In our individual country regression analysis, we faced a significant reduction in sample size, with data available for only 3 to 4 years in some cases, leading us to omit variables with the most missing data. Although most coefficients had p-values above 0.05, graphical analysis revealed a noticeable trend: lagged official aid appeared to align with the IQI index in most countries. This was particularly evident in Bosnia and Herzegovina, where significant fluctuations in foreign aid correlated with substantial year-to-year changes in the IQI index, although no causal relationship could be definitively established.